Volatility Linkages in the Finnish Stock, Bond, and Money Markets

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Abstract

This paper uses weekly data for the period January 9, 1991 to December 30, 2003 to assert the strength of volatility linkages between Finnish stock, bond, and money markets. We use both generalized method of moments and a vector-autoregressive EGARCH specification. The volatility linkages for the stock–bond, and stock–money market pairings are surprisingly low, even negative, and weaker than the return correlations. Logically, the bond–money market pairing exhibit a strong, yet not perfect volatility link. Stock market has the most 'independent' volatility, influencing both the bond volatility, and the money market volatility. Then again, money market volatility is negatively affected both by the stock market, and the bond market component. The leverage hypothesis for asymmetric volatility is contradicted by significant asymmetry parameters both for the stock market, and the money market. The return correlations between the stock market and the other markets have decreased over time. We conclude that the asset allocation, risk management, and regulatory need for considering linkages beyond return correlations is fairly weak.

Key words: GMM, EGARCH, volatility linkages.

JEL Classification: .C32, G10, G12.

Acknowledgments

I would like to thank my thesis supervisors, professors Eva Liljeblom, Anders Löflund, and Timo Teräsvirta, and Mika Vaihekoski for helpful comments. We also thank Matti Pajari at Nordea Bank for providing data on the Nordea bond and money market indices. The usual disclaimer applies.

This version: January 15, 2004. EFMA 2004, Basel, Switzerland.

1 Introduction

One of the most central decisions an investor has to take is the proportion of his funds to be invested in stock, bonds, and money market instruments. A portfolio manager shifts funds from interest bearing instruments into stocks when he expects higher stock market returns, and vice versa. Similarly, and all else equal, when the expected volatility in one market increases, there is a shift of funds towards the other markets. However, if asset prices are driven by common underlying factors, a shock to one asset might have implications on the movements of other asset classes, or implications back on the fundamentals. Hence prices will not be driven only by own shocks, but also by movements in other assets. For example, while a positive shock to equity prices tends to increase interest rates, the higher rates reduce the present value of future dividend payments. The risk reduction due to shifting investments from one asset category to another is highly due to the return and volatility linkages between the markets. Failing to account for these linkages might have severe implications on portfolio and investment management, and risk allocation. To ensure the adequacy of capital in, for example, the banking industry, also regulators should be interested in the nature of market linkages. Consequently, analyzing only one asset class (henceforth 'market') in isolation leaves important information unused.

There is a vast body of literature on the linkages between financial markets, both first moment – prices and/or returns – and second moment linkages. The general notion is that deregulation, technical achievements, and internationalization should have made financial markets more interdependent. First moment linkages between stock markets are studied for example by Eun and Shim (1989), Kasa (1992), Arshanapalli and Doukas (1993), Corhay et al. (1993), Karolyi and Stulz (1996), Engsted and Lund (1997), and Ahlgren and Antell (2002). Price linkages between Scandinavian markets are studied by Mathur and Subrahmanyam (1990), and Malkamäki et al. (1993). Many researchers have found that the linkages between international markets have increased, especially after the 1987 stock market crash, and during periods of high volatility, i.e., 'crisis periods'.

Second moment linkages between geographic markets are studied, among others, by Hamao et al. (1990), Koutmos and Booth (1995), and Koutmos (1996, 1999). Using a multivariate EGARCH model, and daily data from May, 1988 to June, 1994, Booth et al.

(1997) find (asymmetric) volatility transmission between the Scandinavian countries. However, the price and volatility effects between these markets are surprisingly low and unsystematic.

While there exists a vast body of literature on modeling first and second moments of returns, and the linkages between geographical markets, there have been much less attention towards the linkages between asset categories. Money market, bond market, and equity market data were used in the seminal paper of Bollerslev et al. (1988). They use a trivariate diagonal Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model to estimate the variances and covariances for US bills, bonds, and stocks for quarterly data over the period 1959(1)–1984(2). They find that the conditional variance-covariance matrix is highly autoregressive, and an important determinant of the risk-premia.

Fleming et al. (1998) develop a trading model for the stock, bond, and money markets where volatility linkages arise, first, due to common information that affects all markets simultaneously, and, second, due to cross-market hedging. The volatility is modeled as a function of an (unfortunately) unobservable information flow. The authors expect both common information and spillover to be important. Their model predicts strong volatility linkages. The empirical model is based on an *AR*(1) log volatility process resembling an Exponential GARCH (EGARCH) model, but estimated via the generalized method of moments (GMM). The data consist of daily return observations on SP500 stock index futures, *T*-bond futures, and *T*-bill futures over the period January 1983–August 1995. The results show that there are very strong, however not perfect, volatility linkages. These linkages are much higher than the return correlations. Further, the linkages have strengthened post-1987. Kim et al. (2001) use daily Australian equity, bond, and money market futures data over the period January 1988–December 1999. The econometric specification is a constant correlation trivariate EGARCH model. They find strong volatility linkages between the markets. There are also asymmetric effects both within markets and between markets.

In this paper we use weekly Finnish stock, bond, and money market return data over the period January 9, 1991 to December 30, 2003 to assess the volatility linkages between the three markets. The strength of the linkages is estimated first by the generalized method of moments, and then by a trivariate VAR–EGARCH specification. Con-

trary to Fleming et al. (1998), the strength of the volatility linkages is low, and sometimes even negative. The exception is the link between the bond and money markets, which is stronger than the traditional return correlation. Money market volatility is most dependent on other markets, while the linkages to the stock market is non-existent.

The remainder of the paper is organized as follows. In Section 2 we present a theoretical trading model for stocks, bonds, and money market instruments. Based on this model, empirical GMM moment conditions are derived. Also the VAR–EGARCH specification is shortly described. Next, the weekly stock, bond, and money market data are presented. In Section 4 the empirical results are given. Finally, Section 5 concludes the paper.

2 The Model and Empirical Formulation

2.1 The Theoretical Model

Tauchen and Pitts (1983) develop a trading model for stock futures. Based on this model, and influenced by Ross (1989), Fleming et al. (1998) develops a speculative trading model. Assuming the market is in equilibrium in the beginning of each time period, all new information generates a round of trading that continues until a new equilibrium is reached. This is due to investors' differing expectations and needs to transfer risk. Therefore a large number of speculators trade with each other until a new equilibrium is reached. The amount of trading depends on the flow of information.

Let $F^s_{t,T}$ be the time t price of a stock futures contract maturing at time T, and let S^s_T be the actual stock price realization at time T. A long futures position generates an expected profit of $E(\pi^s_{t,T}) = E(S^s_T) - F^s_{t,T}$, conditioned on available information. Assuming the trader maximizes his expected profit subject to a variance constraint, all standard results from mean-variance analysis apply. The demand function of each trader is expressed as

$$Q_s = \frac{E(\pi^s_{t,T})}{2\alpha\sigma^2_s}, \tag{1}$$

where Q_s is the demanded number of stock futures contracts, α the risk aversion parameter, and σ^2_s the variance of $E(\pi^s_{t,T})$. The number of contracts demanded is related to the expected profit, and inversely related to risk aversion and profit variance.

Fleming et al. (1998) generalizes the above model to allow traders to act on more than one futures market. Take for example an economy where trading occurs both in stock and bond futures. Again, assuming standard mean-variance results, and letting β_s (β_b) denote the slope coefficient in a linear regression of the stock (bond) futures profit $\pi^s_{t,T}$ ($\pi^b_{t,T}$) on the corresponding bond (stock) futures profit $\pi^b_{t,T}$ ($\pi^s_{t,T}$), and $\sigma^2_{s|b}$ ($\sigma^2_{b|s}$) the variance of the regression error, gives the following demand functions:

$$Q_s = \frac{E(\pi^s_{t,T})}{2\alpha\sigma^2_{s|b}} - \frac{E(\pi^b_{t,T})}{2\alpha\sigma^2_{b|s}}\beta_b \tag{2a}$$

for the stock futures contracts, and

$$Q_b = \frac{E(\pi^b_{t,T})}{2\alpha\sigma^2_{b|s}} - \frac{E(\pi^s_{t,T})}{2\alpha\sigma^2_{s|b}}\beta_s \tag{2b}$$

for the bond futures contracts. Now suppose that $E(\pi^b_{t,T}) = 0$. In this case the demand for stock futures is proportionally related to $E(\pi^s_{t,T})$, and inversely related to risk aversion. However, the demand is still related to $\sigma^2_{s|b}$, which means the demand also depends on the correlation between the profits on the stock and bond markets. It is easy to see that the investor's demand for bond futures is $Q_b = -\beta_s Q_s$. The investor is thus willing to take a position in bond futures even when their expected profit is zero. With this hedge in place, the risk of the stock futures position is $\sigma^2_{s|b}$. We have here presented the model in terms of futures contracts. However, through the put-call-futures no-arbitrage parity the trading model applies for spot markets as well.

If the expected profits on both markets are non-zero, the total demand for the contracts is the sum of the investor's speculative and hedging demands. The flow of new information therefore creates market linkages arising from two sources. First, *common information* creates due to changes in speculative demand simultaneous trading on all markets. Second, market specific information causes trading on other markets through changes in hedging demand. This is called *information spillover*. In order to incur hedging benefits, investors diversify across markets. Since market specific information through changes in hedging demand also affects the other markets, the model predicts that volatility changes across frictionless markets are perfectly correlated. Since markets are not frictionless, there are transaction costs, and investors may be restricted by

trading constraints, the information spillover effects across markets will in practice not be perfect. The higher levels of common information and information spillover, the stronger are the linkages. Common information is pronounced in markets that are driven by the same underlying fundamentals, and information spillover should be high between markets where the benefits of cross-market hedging are high.

2.2 Empirical Operationalization

For the empirical formulation of the trading model, let $R_{j,t+1}$ be the logarithmic return in excess of the risk free interest rate for market j, j = {stock, bond, and money market} at time t+1. During each trading period there is a flow of new information, which causes unpredictable price changes. Further, let $\mu_{j,t+1}$ be the conditionally expected return on market j, $\varepsilon_{ij,t+1}$, the i.i.d. normal random variables with mean zero and variance $\sigma^2_{\varepsilon,j}$, i = 1, …, $I_{j,t+1}$, and $I_{j,t+1}$ the number of unobservable information events affecting market j during a unit of time. The return generating process is the sum of the conditionally expected return, and the shocks generated by a large number of unobservable daily information events:

$$R_{j,t+1} = \mu_{j,t+1} + \sum_{i=1}^{I_{j,t+1}} \varepsilon_{ij,t+1} . \tag{3}$$

We can rewrite (3) as

$$R_{j,t+1} = \mu_{j,t+1} + \sigma_{\varepsilon,j} I_{j,t+1}^{½} z_{j,t+1}, \tag{4}$$

where

$$z_{j,t+1} = \frac{1}{I_{j,t+1}^{½}} \sum_{i=1}^{I_{j,t+1}} \frac{\varepsilon_{ij,t+1}}{\sigma_{\varepsilon,j}} . \tag{5}$$

The central limit theorem implies that $z_{j,t+1} \xrightarrow{d} N(0,1)$ as $I_{j,t+1} \to \infty$, and the conditionally expected value of $R_{j,t+1}$ is $\mu_{j,t+1}$ with variance $\sigma^2_{\varepsilon,j} I_{j,t+1}$, and volatility $\sigma_{\varepsilon,j} I_{j,t+1}^{½}$. Thus, the variance of the return is proportional to the number of information events during the period. This assumption resembles Ross (1989).

Next, also the variance should be modeled. A large body of research on financial market volatility suggests an autoregressive structure of the variance should be appropriate. See for example Bollerslev et al. (1993, 1994). Define the time t+1 return variance as

$$h_{j,t+1} = \ln(\sigma^2_{\varepsilon,j} I_{j,t+1}) = \ln(\sigma^2_{\varepsilon,j}) + \ln(I_{j,t+1}) \Leftrightarrow \exp(½ h_{j,t+1}) = \sigma_{\varepsilon,j} I_{j,t+1}^{½} . \tag{6}$$

Combining equations (4) and (6) and parameterizing the variance as a parsimonious AR(1) process gives the joint return generating process

$$R_{j,t+1} = \mu_{j,t+1} + \exp(½h_{j,t+1})z_{j,t+1}, \tag{7a}$$

$$h_{j,t+1} = \gamma_{h,j} + \phi_{h,j}h_{j,t} + u_{j,t+1}, \tag{7b}$$

where $u_{j,t+1}$ is i.i.d. with zero mean, variance $\sigma^2_{u,j}$, and independent of $z_{j,t+1}$. Equation (7b) can be seen as a stochastic volatility analogue to EGARCH.

By convenience, define the unpredictable part of (7a) as

$$r_{j,t+1} = R_{j,t+1} - \mu_{j,t+1}. \tag{8}$$

$r_{j,t+1}$ is stationary by definition. However, since there might be higher-order serial dependence, we define the following logarithm of the squared innovation as

$$\ln r^2_{j,t+1} = h_{j,t+1} + \ln z^2_{j,t+1}. \tag{9}$$

It can be shown that the mean and variance of $\ln z^2_{j,t+1}$ are –1.27, and 4.93, respectively. Subtracting the expected value of $\ln z^2_{j,t+1}$ from $\ln r^2_{j,t+1}$, and denoting it by $y_{j,t+1}$, gives the final and empirically more suitable development of the model,

$$y_{j,t+1} = \ln r^2_{j,t+1} - E(\ln z^2_{j,t+1}) = h_{j,t+1} + \eta_{j,t+1}, \tag{10a}$$

$$h_{j,t+1} = \gamma_{h,j} + \phi_{h,j}h_{j,t} + u_{j,t+1}, \tag{10b}$$

where $\eta_{j,t+1}$ is the mean zero, with variance 4.93 demeaned value of $\ln z^2_{j,t+1}$. Equations (10a) and (10b) with their underlying assumptions imply

$$E[y_{j,t+1}] = E[h_{j,t+1}], \tag{11a}$$

$$\mathrm{var}[y_{j,t+1}] = \mathrm{var}[h_{j,t+1}] + \mathrm{var}[\eta_{j,t+1}], \tag{11b}$$

$$\mathrm{cov}[y_{j,t+1}, y_{j,t+1+\tau}] = (\phi_{h,j})^{\tau}\mathrm{var}[h_{j,t+1}]. \tag{11c}$$

$\tau>0$ is an integer. The autocovariance function in (11c) is identical to that of an ARIMA(1,1) process. The variance can be decomposed according to equation (6) as $h_{j,t+1} = \ln(\sigma^2_{\varepsilon,j}) + \ln(I_{j,t+1})$. Since $\ln(\sigma^2_{\varepsilon,j})$ is a constant, its variance is zero. Since it also is independent of $\ln(I_{j,t+1})$, what we have left is $\mathrm{var}[\ln(I_{j,t+1})]$. In other words, the variance is modeled as a function of log information flows.

Up till now we have dealt only with a single market, *j*. To assess the strength of cross-market volatility linkages, consider another market *k*. If the individual markets *j* and *k* follow the same type of process as in (10a) and (10b), it can easily be shown that cross-market linkages obtain the following form:

$$\text{cov}[y_{j,t+1}, y_{k,t+1}] = \text{cov}[h_{j,t+1}, h_{k,t+1}] + \text{cov}[\eta_{j,t+1}, \eta_{k,t+1}], \tag{12a}$$

$$\text{cov}[y_{j,t+1}, y_{k,t+1+\tau}] = (\phi_{h,k})^{\tau}\text{cov}[h_{j,t+1}, h_{k,t+1}], \tag{12b}$$

$$\text{cov}[y_{j,t+1+\tau}, y_{k,t+1}] = (\phi_{h,j})^{\tau}\text{cov}[h_{j,t+1}, h_{k,t+1}]. \tag{12c}$$

Since the variance of individual markets is modeled as a function of information flows, the correlation between $h_{j,t+1}$ and $h_{k,t+1}$ is a measure of the strength of the informational linkages between markets j and k. The restrictions implied by systems (11) and (12) can be tested by the GMM.

2.3 Estimation Issues

The implications of the theoretical setting in equations (11a)–(11c) in the univariate case, added with equation equations (12a)–(12c) in the bivariate case, can be transformed to an empirically testable GMM implementation. Consider the bivariate model, with two markets, j and k. Equations (11a)–(11c) and (12a)–(12c) imply the following GMM moment conditions, by which the adequacy of the model can be tested:

$$e_{1,t+1} = y_{j,t+1} - \mu_{h,j}, \tag{13a}$$

$$e_{2,t+1} = (y_{j,t+1} - \mu_{h,j})^2 - \sigma^2_{h,j} - 4.93, \tag{13b}$$

$$e_{3,t+1} = (y_{j,t+1} - \mu_{h,j})\,(y_{j,t+1+\tau} - \mu_{h,j}) - (\phi_{h,j})^{\tau}[(y_{j,t+1} - \mu_{h,j})^2 - 4.93], \tag{13c}$$

$$e_{4,t+1} = y_{k,t+1} - \mu_{h,k}, \tag{13d}$$

$$e_{5,t+1} = (y_{k,t+1} - \mu_{h,k})^2 - \sigma^2_{h,k} - 4.93, \tag{13e}$$

$$e_{6,t+1} = (y_{k,t+1} - \mu_{h,k})\,(y_{k,t+1+\tau} - \mu_{h,k}) - (\phi_{h,k})^{\tau}[(y_{k,t+1} - \mu_{h,k})^2 - 4.93], \tag{13f}$$

$$e_{7,t+1} = (y_{j,t+1} - \mu_{h,j})\,(y_{k,t+1} - \mu_{h,k}) - \rho_{h,jk}\sigma_{h,j}\sigma_{h,k} - 4.93\rho_{\eta,jk}, \tag{13g}$$

$$e_{8,t+1} = (y_{j,t+1} - \mu_{h,j})\,(y_{k,t+1+\tau} - \mu_{h,k}) - (\phi_{h,k})^{\tau}[(y_{j,t+1} - \mu_{h,j})\,(y_{k,t+1} - \mu_{h,k}) - 4.93\rho_{\eta,jk}], \tag{13h}$$

$$e_{9,t+1} = (y_{j,t+1+\tau} - \mu_{h,j})\,(y_{k,t+1} - \mu_{h,k}) - (\phi_{h,j})^{\tau}[(y_{j,t+1} - \mu_{h,j})\,(y_{k,t+1} - \mu_{h,k}) - 4.93\rho_{\eta,jk}]. \tag{13i}$$

Equations (13a)–(13c), and (13d)–(13f) relate to the univariate situation for the both markets j and k, respectively, while equations (13g)–(13i) connect the two markets, testing the degree of cross-market linkages. The disturbances in (13) are stacked to the GMM vector $\mathbf{e}_{t+1}(\theta) = [e_{1,t+1}, \ldots, e_{9,t+1}]$. The 8×1 parameter vector is $\theta = [\mu_{h,j}, \sigma^2_{h,j}, \phi_{h,j}, \mu_{h,k}, \sigma^2_{h,k}, \phi_{h,k}, \rho_{h,jk}, \rho_{\eta,jk}]$, and $\tau = 1,2, \ldots, \ell$ is the lag length in the autocovariance function. There are thus q = eight unknown parameters, and $r = 4\times\ell+5$ GMM moment restrictions, leaving $r–q = 4\times\ell–3$ overidentifying restrictions for the GMM test.

The GMM parameter estimates are obtained by minimizing the norm $J(\theta) = \boldsymbol{g}_t(\theta)'\boldsymbol{W}\boldsymbol{g}_t(\theta)$, where $\boldsymbol{g}_t(\theta)$ is a vector-valued function of sample moment conditions[1], and $\boldsymbol{W}$ the weighting matrix. It can be shown that the optimal weighting matrix is the inverse of the variance-covariance matrix, which is corrected for autocorrelated and heteroskedastic disturbances according to Newey and West (1987a). If there are equally many moment conditions (r) and parameters to estimate (q), the system is exactly identified and $J(\theta)$ is zero. However, if there are more moment conditions than parameters, $Q = TJ(\theta) \sim \chi^2(r-q)$ is a test of the overidentified restrictions and is therefore a test of the restrictions imposed by the trading model. Nested models can be can be tested with the GMM counterparts to the LM, LR, and Wald statistics of Newey and West (1987b).

The GMM representation rests on the theoretical trading model outlined earlier in this section. However, a large body of research has found evidence of volatility patterns that conform to the GARCH family of models. Since the autoregressive logarithmic variance parameterization in our GMM specification resembles that of the Exponential GARCH of Nelson (1991), we also conduct the estimations using this method. The following trivariate constant correlation VAR(1)-EGARCH model is similar to Koutmos (1996), allowing an asymmetric response of innovations on variance. For j,k = {stock (S), bond (B), money (M)}, let[2]

$$R_{j,t+1} = \alpha_{j0} + \alpha_{jS}R_{S,t} + \alpha_{jB}R_{B,t} + \alpha_{jM}R_{M,t} + e_{j,t+1},\ e_{j,t+1} \sim N(0, h_{jj,t+1}), \tag{14a}$$

$$h_{jj,t+1} = \exp[\beta_{j0} + \beta_{jS}f_{S,t} + \beta_{jB}f_{B,t} + \beta_{jM}f_{M,t} + \gamma_j \ln(h_{jj,t})], \tag{14b}$$

$$h_{jk,t+1} = \rho_{jk}\sqrt{h_{jj,t+1}}\sqrt{h_{kk,t+1}}\ , \tag{14c}$$

where

$$f_{j,t} = \left|z_{j,t}\right| - E\left|z_{j,t}\right| + \delta_j z_{j,t},\ z_{j,t} = \frac{e_{j,t}}{\sqrt{h_{jj,t}}}, \tag{14d}$$

where $R_{j,t+1}$ follows the previous notation and stands for the excess return on index j = {stock, bond, money}, and is in the mean equation (14a) modeled as a vector autoregressive VAR(1) process of own lagged returns and cross-market returns. The conditional variance $h_{jj,t+1}$ in equation (14b) is specified as a function of own and cross-market past standardized innovations, and own past logarithmic variance. The covariance terms in equation (14c) are based on a constant correlation assumption. The

[1] In the specifications of GMM system (13), $\boldsymbol{g}_t(\theta)$ is simply the sample average of $\boldsymbol{e}_{t+1}(\theta)$.
[2] We partly use the same denotation as before but in a new meaning. This is done only if there is no risk of confusion.

specification in (14) allows both own and cross-market returns to affect future returns. The variance is also predicted by own and cross-market innovations, and own variance. $|z_{j,t}| - E|z_{j,t}|$ in equation (14d) can be thought as magnitude effect, while $\delta_j z_{j,t}$ is a measure of asymmetry. If δ_j is negative, a negative innovation tends to increase volatility more than a positive shock of equal magnitude. Further, the cross-market effect on variance is asymmetric if a positive β_{jk} is combined with a negative δ_j.[3]

Define $\boldsymbol{e_{t+1}} = [e_{stock,t+1} \quad e_{bond,t+1} \quad e_{money,t+1}]'$ to be the mean zero 3×1 disturbance vector. The variance-covariance matrix of $\boldsymbol{e_t}$ is $\boldsymbol{H_t}$ (3×3), defined by the elements in equations (14b) and (14c). To keep the estimation process tractable, we assume a trivariate conditional normal distribution. Its log likelihood value, omitting the constant, at time t is

$$\ln L_t = -\frac{1}{2}\ln|\boldsymbol{H_t}| - \frac{1}{2} e_t' \boldsymbol{H_t^{-1}} \boldsymbol{e_t} . \tag{15}$$

The estimation is conducted by using standard maximum likelihood procedures. To get reliable initial parameter values, we start by estimating univariate specifications, then a bivariate with all cross-terms set to zero, after which also cross-terms are considered. Finally, the trivariate model is estimated similarly with zero cross-terms, and then with non-zero terms. For all estimations initial parameter values are obtained by a simplex algorithm, while the Berndt–Hall–Hall–Hausman (1974) algorithm is used for the actual estimation.

3 The Data

The trading model developed in Fleming et al. (1998) is based on the use of futures contracts. Due to the lack of futures data for the Finnish stock, bond, and money markets, we use spot prices instead. However, due to the no-arbitrage pricing relation between the spot and futures markets, this is not too restrictive. The drawback is that it is harder to go short in spot assets than asset derivatives.

We use two different measures as approximations for the stock market return. The first is the total return HEX General Yield Index, and the second the HEX Portfolio Yield Index, which does not permit the weight of any stock to exceed ten per cent. Both indices account for dividends, splits and alike. HEX Portfolio is motivated by the fact that the

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[3] Asymmetry is modelled in equation (15d). The partial derivatives with respect to z_j are $\partial f_j(z_{j,t+1})/\partial z_{j,t+1} = 1+\delta_j$ for $z_j > 0$, and $\partial f_j(z_{j,t+1})/\partial z_{j,t+1} = -1+\delta_j$ for $z_j < 0$. Therefore a significantly

weight of the largest constituent, Nokia Corporation, for long times has been higher than 50 per cent. The bond market return is approximated the Nordea Government Bond Index, while the money market return is based on the Nordea Money Market Index. The former consists of Finnish government benchmark bonds with a remaining maturity larger than one year. The latter includes money market instruments with a maturity of 1–12 months (average 183 days). All returns are calculated as logarithmic changes of the indices, and are in excess of the holding-period return of the one-month Euribor (prior to 1999 the Helibor). The currency of denomination is the Finnish Markka up to 1999, after which it is the Euro. The data consist of 678 weekly observations (Wednesdays) over the period January 9, 1991 to December 30, 2003. The data period is restricted by the availability of data. Days when any of the markets have been closed are excluded from the data set. Weekly data ensures that the impact of different time stamps for the different markets decreases.

To calculate basic descriptive statistics, we use the GMM based methodology outlined in Richardson and Smith (1993), and used in Harvey (1995), Bekaert and Harvey (1997), and for Finnish data in Antell (2000). Generally, let x_{it} be the realization of a variable. To obtain the mean, variance, skewness, and excess kurtosis, we estimate the following exactly identified GMM system:

$$u_{1it} = x_{it} - \mu_i, \tag{16a}$$

$$u_{2it} = (x_{it} - \mu_i)^2 - \sigma_i^2, \tag{16b}$$

$$u_{3it} = (x_{it} - \mu_i)^3/\sigma_i^3 - skew_i, \tag{16c}$$

$$u_{4it} = (x_{it} - \mu_i)^4/\sigma_i^4 - 3 - xkurt_i, \tag{16d}$$

where μ, σ, *skew*, and *xkurt* depict the mean, standard deviation, skewness, and excess kurtosis, respectively. Normality can be tested by restricting $skew_i$ and $xkurt_i$ to zero, and calculating the GMM overidentifying Q statistic. Alternatively, following Newey and West (1987b), normality can be tested with the Wald for GMM test statistic. Both statistics are distributed as $\chi^2(2)$.

Table 1 shows the descriptive statistics. The weekly mean stock market excess return is 0.20 per cent. On an annualized basis this corresponds roughly to 10.5 per cent. The weekly standard deviation is 4.59 per cent (annualized value 33.13 per cent). The volatility has been clearly higher during the latter part of the data period. The skewness is

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negative δ_j has a larger impact than a positive if $z_{j,t+1}$ is negative.

insignificantly negative, while the excess kurtosis is significantly positive. Both the Wald for GMM and GMM overidentifying statistics reject normality. The annualized return of HEX Portfolio is 5.59 per cent with a standard deviation of 23.2 per cent. The lower values compared to the unrestricted HEX General index is a clear indication of the impact of Nokia. The average annualized bond excess return and volatility are 3.5 per cent, and 4.9 per cent, respectively. The skewness is insignificantly negative, and the excess kurtosis clearly positive. The money market annualized excess return is 0.36 per cent, and fairly risk-free. The skewness and excess kurtosis are positive but insignificant. Also the Wald normality statistics are insignificant both for bond returns as for money market returns. However, the GMM based *Q* statistic rejects normality. The returns and volatilities are, as expected, highest for the stock market and lowest for the money market.

Next, Table 1 shows the first-, second- and third-order sample autocorrelation coefficients. They are surprisingly high, especially for the money market returns. While the individual autocorrelation coefficients give a measure for a given lag order, the LB(·) statistics test for autocorrelation up to a given order.[4] The LB(5) and LB(10) indicate that there exist some first-moment temporal dependencies. The $LB^2(\cdot)$ statistic measures the existence of second-moment nonlinear dependencies. Since the $LB^2(\cdot)$ values are higher than the LB(·) statistics, this indicates that second-moment nonlinear dependencies are more pronounced than first-order dependencies. This is generally the result for most financial market time-series.

To give some preliminary insights to the volatility linkages, Table 2 presents the unconditional correlation coefficients between five different volatility measures for the three markets. The first measure is the absolute excess return, $|R|$. Next, we consider the squared return, and the ten-week rolling standard deviation. Also GARCH–GJR and EGARCH estimates are calculated. The former is the specification developed by Glosten, Jagannathan and Runkle (1993), and the latter by Nelson (1991). Both GARCH models allow an asymmetric response of negative shocks on volatility. We assume normally distributed innovations. (However, the correlation results were the same also for the *t* distribution and the generalized error distribution.)

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[4] The LB (Ljung–Box, 1978) Portmanteau statistics are defined as $\mathrm{LB}(s) = T(T+2)\sum_{i=1}^{s} \frac{\rho_i^2}{T-i}$, where *s* is the number of lags, and *T* the number of observations. LB(*s*) is distributed as $\chi^2(s)$.

The first row of Table 2 shows the return correlations. Let us first consider the correlations with HEX General as measure for the stock market. The correlation between the stock and bond markets, and between the stock and money markets is fairly low, 0.08 and 0.10. The corresponding correlation between the bond and money markets is higher, 0.52. Also for the different volatility measures the correlation is highest between the bond and money markets, ranging from 0.24 for squared returns to 0.41 for the absolute returns. The volatility correlation between the stock market and the bond, and money markets is low, and usually negative, being especially pronounced for the money market. Thus, periods of high volatility in the stock market is countered by lower volatility in the money market. This finding clearly contradicts the positive volatility correlations of Fleming et al. (1998). The return correlations are clearly higher than the volatility correlations for all three markets. However, the situation is different if HEX Portfolio is used as stock market measure. First, the correlations are always higher than for HEX General. Second, all correlations are positive. Third, while many of the correlations using HEX General are negative, they have now turned positive.

4 Empirical Findings

To conduct the GMM estimation, we need the empirical $y_{j,t+1}$, j = stock, bond, and money market, series. They are obtained as follows. It is vital to get mean zero return series $r_{j,t+1}$ as in equation (8). Therefore the excess returns $R_{j,t+1}$ are regressed on a constant, a dummy variable obtaining value one 1993 onwards ($d93$), and another indicator variable taking value one 1999 onwards ($d99$). From the beginning of 1993 foreigners were allowed to freely invest in Finnish securities, and from the beginning of 1999 the trading currency of denomination for Finnish securities changed from Finnish Markka to the Euro. The price formation might have changed due to these events. The choice of variables is fairly ad hoc. However, they are used just to obtain the mean zero series $r_{j,t+1}$. Next, we take the natural logarithm of the squared $r_{j,t+1}$ series. Finally, the series used in the GMM estimation procedure, $y_{j,t+1}$, is calculated by subtracting $E(\ln z_{j,t+1}^2)$ =–1.27 from the natural logarithm of the squared $r_{j,t+1}$ series.

Panel A of Table 3 gives some basic descriptive statistics for $y_{j,t+1}$, obtained in the same way as the statistics in Section 3. The mean is highest for the series of HEX General, and lowest for the money market. Since $y_{j,t+1}$ is a measure of the log variance $h_{j,t+1} = \ln(\sigma_{\varepsilon,j}^2 I_{j,t+1})$, its mean value can be transformed back to ‘standard variance’ as

$\exp(\bar{y}_{j,t+1})$, and to volatility as $\exp(0.5 \times \bar{y}_{j,t+1})$. Further transforming this to yearly standard deviation yields return standard deviations of 27.3 per cent for HEX General (compared to 33.1 for the return standard deviation shown in Table 1) to 0.29 per cent for money market (0.36). The skewnesses are always negative and significant, and the excess kurtosis positive and insignificant only for money market. Normality is always rejected. Also the Ljung–Box (1978) statistics reject the null of no autocorrelation for all but HEX Portfolio. Panel B shows the correlation coefficients. Logically, the correlation is highest between the two measures of stock market returns, and between the bond and money markets. Finally, Figure 1 shows that the series are fairly variable. This can also be seen in the high standard deviation in Panel A of Table 3.

To conduct the GMM estimation of equations (13a)–(13i), the autocovariance order must be defined. Many studies suggest that the volatility is a slowly decaying function of past volatilities. This is also indicated by the $LB^2(\cdot)$ statistics in Table 1. Therefore a large ℓ should be chosen. However, the higher ℓ, the more GMM moment restrictions there will be. Leaving the number of observations constant, it is a well-known fact that the more moment restrictions, the more unstable the GMM system becomes. Therefore we choose ℓ = 20. (The estimations were actually conducted for all autocovariance orders from one to fifty, and approximately from ℓ = 15 upwards the results remained materially the same.)

As a special case, the evaluation is started by estimating univariate specifications for each market, i.e., estimating equations (13a)–(13c). In this case the dimension of the parameter vector $\boldsymbol{\theta}$ is q = 3, and the number of orthogonality conditions $r = \ell+2$, leaving $r–q = \ell–1$ overidentifying restrictions. With ℓ = 20, r = 22, and $r–q$ = 19. The corresponding values for the bivariate specifications are r = 85, and $r–q$ = 77, respectively.

Panel A of Table 4 shows the GMM results for the univariate models. The mean of $h_{j,t+1}$ for HEX General is –6.51 (corresponding to an annualized standard deviation of 27.84 per cent). The value for HEX Portfolio is –7.06 (annualized 21.11 per cent), for the bond market –10.72 (3.40 per cent), and for money market –15.54 (0.30 per cent). These values are logical and in line with the descriptives in Tables 1 and 3. The variance of $h_{j,t+1}$ is highest for the money market, a result obtained also in Fleming et al. (1998). The autocorrelation coefficients $\phi_{h,j}$ are fairly high, close to one, indicating that

the process is very slowly decaying. However, and most importantly, they do not exceed one, in which case the process would be explosive. The coefficient is highest for HEX Portfolio, 0.9919, and lowest for the bond market, 0.8410. However, the coefficient is significantly smaller than one only for the bond, and money markets. 11 of the 12 parameters are significant. The GMM *Q* statistics do not reject the null hypothesis derived from the trading model. Hence the econometric formulation is adequate to describe volatility linkages.

Next, Panel B of Table 4 shows the results for the bivariate specifications. Let us start with the HEX General index as stock market measure. The parameter estimates are fairly in accordance with those reported in Panel A. The overidentifying *Q* statistics show no evidence of misspecification. The parameter of interest in Panel B is the correlation of log information flows, $\rho_{h,jk}$. Interestingly, and contrary to Fleming et al. (1998), the correlation is very different for the stock–bond market, and stock–money market pairings, being insignificantly negative (–0.089), and significantly negative (–0.378), respectively. Both correlations are also clearly lower than the return correlations shown in Table 2, and somewhat smaller than the correlations for the different volatility measures. However, the correlation for the bond–money market pairing is in line to that of Fleming et al., 0.561 and significant, and larger than the return correlation of 0.52. Considering HEX Portfolio as stock market measure yields slightly different results. The stock–bond, and stock–money market correlations are now positive, but insignificant, and slightly larger than the return correlation. It seems that restricting the weight of stocks results in a higher GMM correlation estimate between log information flows. While Fleming et al. (1998) obtain very strong, yet not perfect, volatility linkages, our GMM specification reveals a much lower degree of linkages and information spillover between the markets. If we reflect on HEX General as stock market measure, the volatility linkage is negative! In light of these results, it does not seem worthwhile for the Finnish investor, nor regulatory authorities, to consider correlation measures beyond traditional return correlations.

Next, we estimate the VAR–EGARCH specification. It is based on the HEX General as stock market measure.[5] The results are shown in Tables 5 and 6. In Model 1 all cross-

[5] We conducted the estimation also for HEX Portfolio as stock market measure. However, the results were fairly in line with the results with HEX General as stock market measure. Therefore we do not report the results here.

terms are set to zero, while Model 2 allows free parameter values. There are twelve more parameters in the unrestricted model than the restricted. Although many of the cross-terms turn out insignificant the likelihood ratio statistic 48.45 clearly rejects the parameter restriction. The same result is obtained by the Wald statistic in Table 6. The only individually significant mean equation cross-parameter is between the bond market and money market, the latter affecting positively the return of the former. 'Own' variance equation parameters in Panel B of Table 5 are always significant, the cross-terms not as often. However, there are three significantly negative cross-terms. Bond volatility is affected by the stock market lagged innovation (coefficient –0.1020). Money market volatility is influenced by the lagged stock market component (coefficient –0.0629), and the bond component (coefficient –0.0624). Hence a positive shock on the stock market lowers bond volatility. Positive stock, and bond innovations both lower money market volatility. Also the Wald statistics in Table 6 show the significance of the cross-terms for the bond market (Wald = 9.17), and the money market (41.48). Hence the volatility process is most 'independent' for the stock market, while money market volatility has the strongest link to the two other markets.

The asymmetry effect, first discovered by Black (1976), and Christie (1982), is often referred to as the leverage effect. A negative shock to equity increases the debt-to-equity ratio, making the stock riskier, hence increasing volatility. Another explanation for this asymmetry is the volatility feedback story provided by Pindyck (1984), French et al. (1987), and Campbell and Hentschel (1992). It states that an expected volatility increase causes an instantaneous negative return shock, contradicting the leverage hypothesis. In Table 5 the asymmetry term is always negative, indicating asymmetry. As expected, it is significant for the stock market. Since it is clearly significant also for the money market, the leverage hypothesis cannot be the sole explanation for the financial market asymmetry observed in a large body of empirical literature.

Finally, the correlation coefficients are around 0.10 for the stock-bond pairing, 0.06 for the stock-money pairing, and 0.62 for the bond-money market pairing. These values are fairly in line with the return correlations in table 2. Since these values are return correlations, they should not be compared to the GMM correlation coefficients in Table 4. However, based on the cross-terms in the variance equations, the EGARCH results conform fairly well to the GMM volatility linkages with negative correlations for the stock–bond, and stock–money pairings, of which the latter significant, and a signifi-

cantly positive bond–money pairing. As noted above, there is indeed a negative relation (coefficient –0.1020) between bond volatility and lagged stock innovations. The same is true for the stock–money combination (coefficient –0.0629). However, the bond–money pairing is also negative, while it was positive in the case of GMM.

To assess the goodness of the model, a set of diagnostic tests is conducted. Let $\hat{z}_{j,t} = \hat{\varepsilon}_{j,t} / h_{j,t}^{1/2}$ be the standardized residual for market j = {stock, bond, money}. If the EGARCH is well-specified, this standardized innovation should have zero mean, skewness and excess kurtosis, and unit variance. Using the GMM procedure outlined in equations (16a)–(16d), and previously implemented in tables 1 and 3, the univariate statistics regarding the mean, variance, and skewness in Panel A of Table 7 show little evidence of misspecification. Of these only the skewness test for the stock market is significant. While the coefficients for the excess kurtosis are noticeably lower than the descriptive coefficients in Table 1, they are still very significant. The EGARCH specification has been able to capture much, yet not all of the kurtosis in the raw data. The same picture can be observed in the GMM joint tests in Panel C. Only the joint kurtosis test is significant, driving the total joint statistic significant as well. One solution would be to use some other distributional assumption than the normal distribution. However, the estimation process would be much more difficult.

Further, Panel A shows the univariate Ljung–Box (1978) statistics. Only two of the first-moment statistics are significant. The sign bias, negative size bias, positive size bias, and the joint tests of Engle and Ng (1993) show little evidence of unmodeled asymmetry. Even though it is customary to present these statistics also for multivariate models, it should be noted that they are developed for univariate models, and strictly speaking not applicable for the multivariate case. The covariance bias tests of Koutmos and Knif (2002) in Panel B show that unmodeled covariance asymmetry is usually not a problem. Finally, extending the univariate moments tests of Bekaert and Harvey (1997), also estimated in Antell (2000), to the trivariate case implies $E[\hat{z}_{S,t}\hat{z}_{S,t-\tau}] = E[\hat{z}_{B,t}\hat{z}_{B,t-\tau}] = E[\hat{z}_{M,t}\hat{z}_{M,t-\tau}] = 0$ (trivariate means test), and $E[(\hat{z}_{S,t}^2 - 1)(\hat{z}_{S,t-\tau}^2 - 1)] = E[(\hat{z}_{B,t}^2 - 1)(\hat{z}_{B,t-\tau}^2 - 1)] = E[(\hat{z}_{M,t}^2 - 1)(\hat{z}_{M,t-\tau}^2 - 1)] = 0$ (trivariate variance test). Here we choose τ = 1,…, 5 for the autocovariance lag length, testing *up to* order 5. The results in Panel C show that there are no misspecification problems according to these statistics.

So far we have treated the correlation coefficient constant in all our estimations. However, as many financial parameters are time-varying, that might be the case for the correlation as well. Therefore we model the correlation as a function of explanatory variables. We use the following simple functional form for the correlation coefficient:

$$\rho_{jk,t} = \rho_{0,jk,t} + \rho_{1,jk,t}\, x_t, \tag{17}$$

where x_t is the explanatory variable. We use three variables: the same two variables as previously, an indicator variable for the period beginning 1993, the year when foreign investors were allowed to freely invest in the Finnish financial markets (*d*93), an indicator onwards from 1999, the first year Finnish securities were traded in euro (*d*99), added with an indicator for the recession years in the beginning of the 1990's (*d*recession). It takes value one from January 1991 to the end of March 1993. We will also use the log of the time trend to measure the general development.

Because a positive definite GMM weighting matrix was unattainable, the GMM correlation coefficient for the strength of volatility linkages was not possible to model in any way. However, the EGARCH return correlation coefficient in equation (14c) could be made time-varying according to (17). Table 8 shows the results. It seems that the correlation of the stock market to the bond market, and the money market has decreased since the beginning of the period, especially after introducing the euro. According to the log likelihood values, and the likelihood ratio test statistics, *d*99 is the single most important variable explaining the correlation coefficient. The development of the bond–money market pairing is the opposite, i.e., the coefficient has grown over the 1990's. Due to correlated indicators, one should be careful interpreting the largest specification including all indicators. Finally, we check the correlation coefficient by including the log trend in the model. It turns out very significant for the stock–bond pairing (negative development), and the bond–money pairing (positive development), and insignificantly negative for the stock–money pairing.

5 Conclusions

In portfolio theory investment decisions are made on the basis of return correlations between assets. However, the optimal allocations might be hard to obtain if there exist linkages beyond return correlations. Assessing the strength of volatility linkages between financial markets has important implications for asset pricing and allocation, portfolio risk management, hedging and, regulation. Fleming et al. (1998) develop a

trading model, and a stochastic volatility GMM representation. They find strong volatility linkages. In this paper we use weekly Finnish data for the stock, bond, and money markets for the period January 9, 1991 to December 30, 2003, and a similar GMM methodology. The volatility is modeled as a function of unobserved information flow. The stock–bond market pairing, and the stock–money market pairing yielded a volatility link lower than the return correlation. The volatility link between the stock market, measured with the HEX General index, and the money market is surprisingly clearly negative. In this case periods with high stock market volatility are countered by periods of lower volatility in the bond and money markets. However, the link between the bond market and the money market is clearly positive. The corresponding correlations using the HEX Portfolio Yield Index as stock market measure yielded positive values, and against the money market roughly the same as the return correlation.

To compare the results, we also use a trivariate VAR–EGARCH specification. While the correlation coefficient in the GMM setting is a measure of volatility linkages, it is a traditional return correlation in the EGARCH setting. However, volatility linkages are measured in EGARCH through the lagged cross-terms in the variance equations. The return linkages are weak. Only money market past returns significantly affect bond market returns. The volatility linkages are stronger. The only market not affected by other markets' components is the stock market, which is able to affect the volatility of both other markets. The bond market is negatively affected by the stock market component. Money market volatility is affected negatively both by the stock market, and the bond market. Interestingly not only the stock market exhibits asymmetric volatility, but also the money market, challenging the leverage hypothesis for asymmetric volatility. Finally, the EGARCH return correlation coefficient decreases over time for the stock market against the other markets, while the strength of linkages increases between the bond and stock markets.

The empirical results do not verify the perfect correlation assertion of the theoretical trading model. The return and volatility linkages between stock, bond, and money markets are fairly weak during the sample period. Many of the significant linkages were negative. This might be due to problems with the model, market frictions, or the data used. Some information may be lost by using bond indices instead of actual bond returns. On the other hand, an index more accurately captures the development of the market as an entity. Due to spot-futures arbitrage relations the use of spot data instead

of futures contracts should have no impact on the empirical outcome, although the practical implementation of, e.g., short-selling is more troublesome without futures contracts.

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Table 1. Descriptive statistics for Finnish stock, bond, and money market returns. Weekly data January 9, 1991–December 30, 2003.

	Stock Market 1	Stock market 2	Bond Market	Money Market
Mean (%)	0.2024	0.1074	0.0678*	0.0069
Yearly mean (%)	10.5270	5.5866	3.5279	0.3600
Std Deviation (%)	4.5941	3.2190**	0.6856**	0.0935**
Yearly Std Dev (%)	33.1282	23.2123	4.9442	0.6744
Skewness	–0.4280	–0.2543	–1.0521	0.3780
Excess Kurtosis	2.9766**	2.0848	9.8173*	26.1571
Wald	32.96**	15.01**	4.50	3.43
Q	59.58**	22.91**	83.93**	155.89**
$\hat{\rho}_1$	–0.0255	0.0582*	–0.0208	0.0799*
$\hat{\rho}_2$	0.0588*	0.0768*	–0.0238	0.1127*
$\hat{\rho}_3$	0.0580*	0.0849*	0.1603*	0.0593*
LB(5)	16.27**	17.66**	19.20**	18.28**
$LB^2(5)$	47.66**	21.61**	107.72**	26.06**
LB(10)	21.72*	23.30**	27.28**	27.07**
$LB^2(10)$	120.33**	65.76**	128.33**	43.50**

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: Stock market 1 refers to the HEX General Yield index, stock market 2 to the HEX Portfolio Yield index, bond market to Nordea Government Bond Index, and money market to Nordea Money Market index. The mean, standard deviation, skewness, and excess kurtosis are obtained from a univariate GMM system as in equations (16a)–(16d). Wald is the Wald for GMM, and *Q* is the overidentifying GMM test for normality, distributed as $\chi^2(2)$. $\hat{\rho}$ is the sample autocorrelation coefficient. $|\hat{\rho}| > 1/\sqrt{2T}$ is regarded as significantly different from zero. LB(*s*) and $LB^2(s)$ are the Ljung–Box tests for autocorrelation up to order *s* for the variable itself and its square, respectively.

Table 2. Correlation coefficients between different volatility measures. Weekly data January 9, 1991– December 30, 2003.

	Stocks 1 vs bonds	Stocks 1 vs money	Bonds vs money	Stocks 2 vs bonds	Stocks 2 vs money
Return, $\rho(R)$	0.0758	0.1022	0.5164	0.1468	0.1569
Absolute return, $\rho(\lvert R\rvert)$	0.0247	–0.0022	0.4050	0.1120	0.1329
Squared return, $\rho(R^2)$	0.0009	0.0226	0.2355	0.0895	0.1302
Rolling volatility (10 w)	–0.0468	–0.1842	0.3990	0.1728	0.1477
GARCH-GJR (normal)	–0.0899	–0.2492	0.3320	0.1071	0.0949
EGARCH (normal)	–0.0966	–0.1837	0.3768	0.1282	0.1975

Notes: Stock market 1 refers to the HEX General Yield index, stock market 2 to the HEX Portfolio Yield index, bond market to Nordea Government Bond Index, and money market to Nordea Money Market index. The first row gives the return correlation. The next five volatility measures are the absolute return, the squared return, the ten-week rolling standard deviation, the generalized autoregressive conditional heteroskedasticity formulation as in Glosten, Jagannathan and Runkle (1993) with normally distributed errors, and finally the exponential GARCH with normally distributed errors.

Table 3. Descriptive statistics for the generated $y_{j,t+1}$ series for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.

	Stock Market 1	Stock market 2	Bond Market	Money Market
Panel A. Descriptive statistics				
Mean (%)	–6.5442**	–7.1297**	–10.7671**	–15.6305**
Impl. yr std dev. (%)	27.3494	20.4084	3.3110	0.2910
Std Deviation (%)	2.3851**	2.3743**	2.6100**	2.7779**
Skewness	–1.3763**	–1.4915**	–1.1280**	–0.6770*
Excess Kurtosis	3.8210**	3.7945**	2.2942**	2.2507
Wald	114.89**	184.23**	133.93**	12.91**
Q	65.19**	95.10**	68.31**	9.93**
$\hat{\rho}_1$	0.1262*	–0.0063	0.1510*	0.2493*
$\hat{\rho}_2$	0.1453*	0.0599*	0.1185*	0.2911*
$\hat{\rho}_3$	0.0863*	0.0254	0.1019*	0.2383*
LB(5)	44.79**	3.76	36.69**	237.24**
LB(10)	79.43**	17.16	43.94**	404.20**
Panel B. Correlations				
	Stock Market 1	Stock market 2	Bond Market	Money Market
Stock Market 1	1			
Stock Market 2	0.6846	1		
Bond market	0.0087	0.0179	1	
Money market	–0.0579	0.0418	0.2970	1

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: The $y_{j,t+1}$ series is obtained by subtracting –1.27 from the natural logarithm of the de-meaned excess returns. See equations (8)–(10). Stock market 1 refers to the HEX General Yield index, stock market 2 to the HEX Portfolio Yield index, bond market to Nordea Government Bond Index, and money market to Nordea Money Market index. The mean, standard deviation, skewness, and excess kurtosis are obtained from a univariate GMM system as in equations (16a)–(16d). Wald is the Wald for GMM, and *Q* is the overidentifying GMM test for normality, distributed as $\chi^2(2)$. $\hat{\rho}$ is the sample autocorrelation coefficient. $|\hat{\rho}| > 1/\sqrt{2T}$ is regarded as significantly different from zero. LB(*s*) is the Ljung–Box test for autocorrelation up to order *s*.

Table 4. GMM estimation results. Weekly data January 9, 1991– December 30, 2003.

	Stock market 1	Stock market 2	Bond market	Money market
Panel A. Univariate models.				
$\mu_{h,j}$	–6.5084**	–7.0626**	–10.7163**	–15.5435**
Std error	(0.0736)	(0.0706)	(0.0891)	(0.0978)
Implied yr. std dev. (%)	27.8438	21.1050	3.3961	0.3039
$\sigma^2_{h,j}$	0.6399**	0.1843	1.0603**	2.1283**
Std error	(0.1095)	(0.0976)	(0.2724)	(0.2032)
$\phi_{h,j}$	0.9826**	0.9919**	0.8410**	0.9779**
Std error	(0.0121)	(0.0103)	(0.0274)	(0.0048)
GMM *Q* stat.	17.49	11.35	13.06	8.35
p-value	[0.5565]	[0.9117]	[0.8357]	[0.9828]

Panel B. Bivariate models.

	Stocks 1 (*j*) vs bonds (*k*)	Stocks 1 (*j*) vs money (*k*)	Bonds (*j*) vs money (*k*)	Stocks 2 (*j*) vs bonds (*k*)	Stocks 2 (*j*) vs money (*k*)
$\mu_{h,j}$	–6.4984**	–6.4901**	–10.6313**	–7.0769**	–7.0566**
Std error	(0.0692)	(0.0675)	(0.0810)	(0.0660)	(0.0660)
Implied yr. std dev. (%)	27.9824	28.0998	3.5436	20.9542	21.1678
$\sigma^2_{h,j}$	0.6245**	0.6410**	0.7702**	0.1112	0.1597
Std error	(0.0985)	(0.0981)	(0.1811)	(0.0896)	(0.0889)
$\phi_{h,j}$	0.9960**	0.9901**	0.9258**	0.9832**	0.9934**
Std error	(0.0094)	(0.0090)	(0.0091)	(0.0088)	(0.0090)
$\mu_{h,k}$	–10.6878**	–15.6193**	–15.5320**	–10.7375**	–15.5859**
Std error	(0.0819)	(0.0916)	(0.0930)	(0.0820)	(0.0906)
Implied yr. std dev. (%)	3.4449	0.2926	0.3057	3.3603	0.2975
$\sigma^2_{h,k}$	0.8181**	1.9651**	2.0371**	0.8475**	1.9507**
Std error	(0.2082)	(0.1875)	(0.1800)	(0.2293)	(0.1748)
$\phi_{h,k}$	0.8997**	0.9760**	0.9770**	0.8772**	0.9855**
Std error	(0.0139)	(0.0046)	(0.0038)	(0.0176)	(0.0040)
$\rho_{h,jk}$	–0.0893	–0.3776**	0.5611**	0.1973	0.2129
Std error	(0.1057)	(0.0623)	(0.0925)	(0.2502)	(0.1226)
$\phi_{\eta,jk}$	0.0408	0.0052	0.3118**	0.0405	0.0481
Std error	(0.0347)	(0.0363)	(0.0419)	(0.0357)	(0.0368)
GMM *Q* stat.	85.15	62.62	66.06	61.36	55.50
p-value	[0.2456]	[0.8821]	[0.8085]	[0.9036]	[0.9694]

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: Stock market 1 refers to the HEX General Yield index, stock market 2 to the HEX Portfolio Yield index, bond market to Nordea Government Bond Index, and money market to Nordea Money Market index. The univariate specification rests on equations (13a)–(13c), and the bivariate specification on equations (14a)–(14i). The table reports parameter values and GMM overidentifying test statistics *Q*. The *Q* statistic is distributed as $\chi^2(19)$ for the univariate models while it is $\chi^2(77)$ for the bivariate specifications. Standard errors are reported in parentheses, *p*-values in brackets.

Table 5. Trivariate VAR–EGARCH estimates for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.

	Stock Market		Bond Market		Money Market	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Log *L*	9,857.50	9,881.73	LR statistic	48.45	*p*-value	<0.0001
Panel A. Mean equation						
$\alpha_{j0}\times 100$	0. 3788*	0.3161*	0. 1079**	0.1094**	0.0082**	0.0074**
	(0.1459)	(0.1513)	(0.0163)	(0.0177)	(0.0012)	(0.0015)
α_{jS}	–0.0059	0.0162		0.0027		–0.0001
	(0.0451)	(0.0483)		(0.0036)		(0.0002)
α_{jB}		–0.0400	–0.0512	–0.0728		0.0004
		(0.2295)	(0.0359)	(0.0415)		(0.0030)
α_{jM}		1.2639		0.9994**	0.0705*	0.1296**
		(1.6139)		(0.2310)	(0.0279)	(0.0420)
Panel B. Variance equation						
β_{j0}	–0.1018*	–0.1982*	–1.3836**	–1.7665**	–0.1039**	–0.0181*
	(0.0485)	(0.0614)	(0.2401)	(0.2882)	(0.0217)	(0.0185)
β_{jS}	0.2154**	0.2681**		–0.1020**		–0.0629**
	(0.0359)	(0.0449)		(0.0388)		(0.0169)
β_{jB}		–0.0324	0.4367**	0.4393**		–0.0624**
		(0.0363)	(0.0326)	(0.0369)		(0.0136)
β_{jM}		0.0055		0.0139	0.1017**	0.0764**
		(0.0177)		(0.0270)	(0.0121)	(0.0104)
γ_j	0.9834**	0.9684**	0.8616**	0.8249**	0.9928**	0.9991**
	(0.0076)	(0.0096)	(0.0236)	(0.0282)	(0.0016)	(0.0013)
δ_j	–0.2188**	–0.1647*	–0.0508	–0.0289	–0.7414**	–1.0674**
	(0.0720)	(0.0710)	(0.0412)	(0.0514)	(0.0819)	(0.1363)
Correlations	Stocks versus bonds		Stocks vs money market		Bonds vs money market	
	0.1104**	0.1035*	0.0644	0.0563	0.6131**	0.6231**
	(0.0421)	(0.0455)	(0.0427)	(0.0434)	(0.0240)	(0.0254)

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: Model 2 is based on equations (14a)–(14d).

$$R_{j,t+1} = \alpha_{j0} + \alpha_{jS}R_{S,t} + \alpha_{jB}R_{B,t} + \alpha_{jM}R_{M,t} + e_{j,t+1},\ e_{j,t+1} \sim N(0, h_{jj,t+1}), \quad (14a)$$

$$h_{jj,t+1} = \exp[\beta_{j0} + \beta_{jS}f_{S,t} + \beta_{jB}f_{B,t} + \beta_{jM}f_{M,t} + \gamma_j \ln(h_{jj,t})], \quad (14b)$$

$$h_{jk,t+1} = \rho_{jk}\sqrt{h_{jj,t+1}}\sqrt{h_{kk,t+1}}, \quad (14c)$$

$$f_{j,t} = |z_{j,t}| - E|z_{j,t}| + \delta_j z_{j,t},\ z_{j,t} = \frac{e_{j,t}}{\sqrt{h_{jj,t}}}. \quad (14d)$$

In model 1 all cross terms are restricted to zero. Standard errors are in parentheses.

Table 6. Wald joint statistics for trivariate VAR–EGARCH estimates for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.

Six cross terms in the mean equations, $\chi^2(6)$	22.44** [<0.0010]
Two cross terms in the mean equation for the stock market, $\chi^2(2)$	0.65 [0.7224]
Two cross terms in the mean equation for the bond market, $\chi^2(2)$	19.28** [<0.0001]
Two cross terms in the mean equation for the money market, $\chi^2(2)$	0.32 [0.8507]
Six cross terms in the variance equations, $\chi^2(6)$	57.99** [<0.0001]
Two cross terms in the variance equation for the stock market, $\chi^2(2)$	0.80 [0.5710]
Two cross terms in the variance equation for the bond market, $\chi^2(2)$	9.17* [0.0102]
Two cross terms in the variance equation for the money market, $\chi^2(2)$	41.48** [<0.0001]
Three asymmetry parameters, $\chi^2(3)$	55.54** [<0.0001]
Twelve cross terms, $\chi^2(12)$	91.40** [<0.0001]

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: Model 2 is based on equations (14a)–(14d).

$$R_{j,t+1} = \alpha_{j0} + \alpha_{jS} R_{S,t} + \alpha_{jB} R_{B,t} + \alpha_{jM} R_{M,t} + e_{j,t+1},\ e_{j,t+1} \sim N(0, h_{jj,t+1}), \quad (14a)$$

$$h_{jj,t+1} = \exp[\beta_{j0} + \beta_{jS} f_{S,t} + \beta_{jB} f_{B,t} + \beta_{jM} f_{M,t} + \gamma_j \ln(h_{jj,t})], \quad (14b)$$

$$h_{jk,t+1} = \rho_{jk} \sqrt{h_{jj,t+1}} \sqrt{h_{kk,t+1}}, \quad (14c)$$

$$f_{j,t} = |z_{j,t}| - E|z_{j,t}| + \delta_j z_{j,t},\ z_{j,t} = \frac{e_{j,t}}{\sqrt{h_{jj,t}}}. \quad (14d)$$

In model 1 all cross terms are restricted to zero. Standard errors are in parentheses.

Table 7. Residual diagnostics for the trivariate VAR–EGARCH model for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.

	Stock Market	Bond Market	Money Market
Panel A. Univariate statistics			
Mean of $\hat{z}_{j,t}$	–0.0195	–0.0566	–0.0259
Variance of $\hat{z}_{j,t} - 1$	–0.0029	0.0133	0.0439
Skewness of $\hat{z}_{j,t}$	–0.3884*	–0.4012	–0.1497
Excess kurtosis of $\hat{z}_{j,t}$	1.3497**	3.7268**	3.1994**
Wald	13.94**	29.56**	43.09**
Q	10.91**	60.68**	73.34**
LB(5)	15.70**	11.22*	4.06
$LB^2(5)$	6.65	5.24	4.83
LB(10)	17.05	13.35	7.68
$LB^2(10)$	16.85	6.49	6.80
Sign bias t-test	0.26	2.01*	–0.13
Negative size bias t-test	1.69	–1.18	–1.07
Positive size bias t-test	–0.01	–3.60**	0.57
Joint $\chi^2(3)$ test	7.82*	16.48**	1.94
Panel B. Covariance specifications	Stock versus bond market	Stock versus money market	Bond versus money market
Covariance sign bias test	3.64	0.30	3.12
Cov. negative size bias	7.96*	8.80*	3.72
Cov. positive size bias	1.53	3.26	7.71*
Covariance joint test	12.17	15.94*	10.03
LB5(z_{jk})	22.11**	19.53**	7.05
LB10(z_{jk})	49.99**	40.55**	12.98
Panel C. GMM joint tests		χ^2 statistic	*p*-value
$E(\hat{z}_{S,t}) = E(\hat{z}_{B,t}) = E(\hat{z}_{M,t}) = 0$, $\chi^2(3)$		2.37	0.4997
$E(\hat{z}^2_{S,t} - 1) = E(\hat{z}^2_{B,t} - 1) = E(\hat{z}^2_{M,t} - 1) = 0$, $\chi^2(3)$		0.24	0.9710
$E(\hat{z}^3_{S,t}) = E(\hat{z}^3_{B,t} = E(\hat{z}^3_{M,t}) = 0$, $\chi^2(3)$		5.60	0.1327
$E(\hat{z}^4_{S,t} - 3) = E(\hat{z}^4_{B,t} - 3) = E(\hat{z}^4_{M,t} - 3) = 0$, $\chi^2(3)$		13.99**	0.0029
$E(\hat{z}_{S,t}\hat{z}_{B,t} - \hat{\rho}_{S,B}) = E(\hat{z}_{S,t}\hat{z}_{M,t} - \hat{\rho}_{S,M}) = E(\hat{z}_{B,t}\hat{\rho}_{M,B} - \hat{\rho}_{B,M}) = 0$		0.20	0.9771
Joint test		85.32**	<0.0000
Trivariate means test, τ = 5, $\chi^2(15)$		24.47	0.0575
Trivariate variance test, τ = 5, $\chi^2(15)$		19.37	0.1974

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: The statistics are based on equations (14a)–(14d).

$$R_{j,t+1} = \alpha_{j0} + \alpha_{jS}R_{S,t} + \alpha_{jB}R_{B,t} + \alpha_{jM}R_{M,t} + e_{j,t+1},\ e_{j,t+1} \sim N(0, h_{jj,t+1}), \quad (14a)$$

$$h_{jj,t+1} = \exp[\beta_{j0} + \beta_{jS}f_{S,t} + \beta_{jB}f_{B,t} + \beta_{jM}f_{M,t} + \gamma_j \ln(h_{jj,t})], \quad (14b)$$

$$h_{jk,t+1} = \rho_{jk}\sqrt{h_{jj,t+1}}\sqrt{h_{kk,t+1}}, \quad (14c)$$

$$f_{j,t} = |z_{j,t}| - E|z_{j,t}| + \delta_j z_{j,t},\ z_{j,t} = \frac{e_{j,t}}{\sqrt{h_{jj,t}}}. \quad (14d)$$

The first six parameters are estimated by the GMM, and described in equations (16a)–(16d). LB(*s*) and $LB^2(s)$ are the Ljung–Box tests for autocorrelation up to order *s* for the variable itself and its square, respectively. The sign bias, negative size bias, positive size bias, and joint tests are derived by Engle and Ng (1993), and the covariance bias tests by Koutmos and Knif (2002).

Table 8. Modeling the correlation coefficient of the trivariate VAR–EGARCH model for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.

	Stock versus bond market	Stock versus money market	Bond versus money market
Correlation specification			
Constant correlation	0.1035*	0.0563	0.6231**
		Log likelihood	9 881.73
Constant	0.3125**	0.1455	0.4883**
*d*93	–0.2522*	–0.1165	0.1627*
LR-test to constant correlation	12.86**	Log likelihood	9 888.16
Constant	0.3196**	0.1801**	0.6055**
*d*99	–0.5392**	–0.3297**	0.0535
LR-test to constant correlation	51.10**	Log likelihood	9 907.27
Constant	0.0494	0.0182	0.6484**
*d*recession	–0.2839**	0.1611	0.1266*
LR-test to constant correlation	13.49**	Log likelihood	9 888.47
Constant	0.3105**	0.1435	0.4861**
*d*93	0.0146	0.0501	0.1605*
*d*99	–0.5463**	–0.3439**	0.0101
LR-test to constant correlation	56.86**	Log likelihood	9 910.15
Constant	0.0934	–0.1974	0.3759
*d*93	0.2210	0.3758	0.2653
*d*99	–0.5356**	–0.3284**	0.0142
*d*recession	0.2178	0.3372	0.1061
LR-test to constant correlation	59.29**	Log likelihood	9 911.37
Constant	1.0245**	0.2063	0.2720*
ln(time)	–0.1681**	–0.0304	0.0656**
LR-test to constant correlation	40.63**	Log likelihood	9 902.04

* Statistically significant at the 5 per cent level. ** Statistically significant at the 1 per cent level.

Notes: The statistics are based on equations (14a)–(14d).

$$R_{j,t+1} = \alpha_{j0} + \alpha_{jS} R_{S,t} + \alpha_{jB} R_{B,t} + \alpha_{jM} R_{M,t} + e_{j,t+1},\ e_{j,t+1} \sim N(0, h_{jj,t+1}), \quad (14a)$$

$$h_{jj,t+1} = \exp[\beta_{j0} + \beta_{jS} f_{S,t} + \beta_{jB} f_{B,t} + \beta_{jM} f_{M,t} + \gamma_j \ln(h_{jj,t})], \quad (14b)$$

$$h_{jk,t+1} = \rho_{jk} \sqrt{h_{jj,t+1}} \sqrt{h_{kk,t+1}}, \quad (14c)$$

$$f_{j,t} = |z_{j,t}| - E|z_{j,t}| + \delta_j z_{j,t},\ z_{j,t} = \frac{e_{j,t}}{\sqrt{h_{jj,t}}}. \quad (14d)$$

$\rho_{jk,t} = \rho_{0,jk,t} + \rho_{1,jk,t}\, x_t$. There are three explanatory indicator variables: *d*93, taking value one from the beginning of 1993, *d*99, taking value one from the beginning of 1999, and *d*recession, taking value one from January 1991 to March 1993.

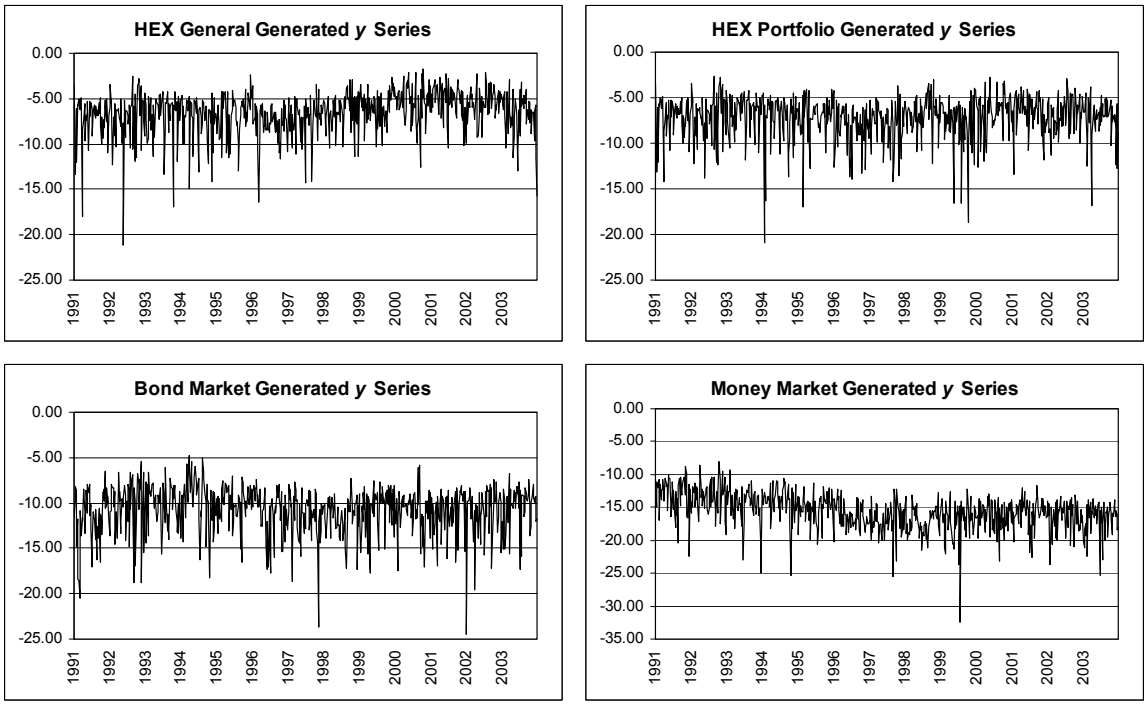

Figure 1. Generated $y_{j,t+1}$ series for Finnish stock, bond, and money market returns. Weekly data January 9, 1991– December 30, 2003.